Gate-Controlled Hole Injection at Top and Edge Contacts to Few-Layer InSe: An Anisotropic, Electro-Thermal TCAD Study Verified Analytically and Benchmarked on a Known Contact

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Abstract

We model hole injection into back-gated, few-layer γ-InSe through top and edge metal contacts with an open-source DEVSIM model. The model couples Poisson's equation to Fermi–Dirac carriers with anisotropic (in-plane/out-of-plane) mobility. Contacts use flux-balanced thermionic emission with a van der Waals (vdW) gap transmission and nonlocal Tsu–Esaki tunnelling along the contact normals. The lattice heat equation has anisotropic conductivity and includes Joule, interface, Peltier and tunnelling heat. Nine verification tests reproduce closed-form results and conservation laws. As a numerical benchmark on a different material, with a single parameter fitted to the measured contact resistance of ultra-high-vacuum Au/MoS_2 contacts (740 $\Omega \cdot \mu$m), the same model predicts the measured transfer length at 300 K (34 vs 35–40 nm) and the measured trends with temperature. For InSe we find: (i) With the Fermi-level pinning measured for direct metal/InSe contacts ($S = 0.05$, pinning level 0.3 eV below E_C) and the measured electron affinity, every common metal gives a hole barrier of 1.05–1.12 eV. Such contacts are electron-injecting, and hole conduction requires gate-induced tunnelling, which carries $\geq$99.9% of the on-current; the on-state resistance then varies by only 2.4$\times$ from Ti to Pt. (ii) At any barrier, edge contacts outperform 50-nm top contacts in the gated on state, and the advantage grows with the barrier: 14$\times$ at 0.2 eV, 51$\times$ at 0.6 eV and 150$\times$ at 1.0 eV. It arises from the vdW gap and from slow out-of-plane transport. (iii) Lowering the hole barrier needs weaker pinning: with the measured oxide interlayer ($S = 0.5$) Pt reaches 0.52 eV and a 700$\times$ lower edge resistance than with direct contacts. The edge advantage disappears only if edge contacts are more strongly pinned than top contacts. (iv) Top contacts have an optimum film thickness of about four layers, and edge contacts improve monotonically with thickness. (v) Edge-contacted devices carry more current and heat correspondingly more; Peltier heat cools the source and heats the drain. A transmission-line compact model reproduces the simulated top-contact resistance with a median error of 11%, and Monte Carlo analysis identifies band alignment and pinning as the dominant uncertainties.

Keywords: indium selenide; Schottky-barrier FET; contact resistance; tunnelling; electro-thermal simulation; van der Waals contacts; compact model; uncertainty quantification

1 Introduction

InSe combines high electron mobility [1], ballistic short-channel transistors [2] and wafer-scale growth [3], but complementary logic also needs efficient hole injection, which has received far less

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attention [4]. Metal contacts to 2D semiconductors are governed by the Schottky barrier and Fermi-level pinning, by the vdW tunnel gap of top-deposited metals, and by how current spreads from the contact into the channel [5, 6]. Edge contacts remove the vdW gap and inject in-plane [7, 8]. Real devices are gated, and the gate can thin the Schottky barrier until tunnelling dominates injection, as in Schottky-barrier FETs. Layered crystals are also strongly anisotropic. Carrier mobility across InSe layers is typically one to three orders of magnitude lower than along them [9], and the through-plane thermal conductivity is about eleven times lower [10].

This paper asks which contact geometry gives the lowest resistance for hole injection into a few-layer InSe p-FET, and why. It also asks how robust that answer is to the interface parameters, and what the measured Fermi-level pinning of InSe contacts [11, 12] implies for p-type operation. We answer with a model in which each of the following ingredients is included and tested:

- a flux-balanced contact condition;
- non-equilibrium Poisson coupling;
- Fermi–Dirac statistics;
- a gate;
- nonlocal tunnelling;
- transport and thermal anisotropy;
- a complete heat source.

We also benchmark its numerics on a well-characterised metal/2D contact (Au/MoS_2) and test its predictions out of sample.

2 Model

2.1 Device and band structure

The device is an N-layer InSe film (0.83 nm per layer; $N = 6$, $t = 5$ nm by default) on 10 nm SiO_2 with a back gate (flat band at $V_G = 0$), with a 50-nm channel. Top contacts of length L_c cover the upper surface, and edge contacts cover the film's end faces. The band gap follows a fit to photoluminescence data [13, 14], $E_g(N) = 1.26 + 2.78N^{-1.70}$ eV, which gives 1.69 eV at 3 layers, 1.39 eV at 6 layers and 1.30 eV at 12 layers. 32% of the confinement shift is assigned to the valence band, roughly $m_e/(m_e+m_h)$. The 6-layer film therefore has $E_V = 5.00$ eV below vacuum (Fig. 9a). All parameters and their provenance are listed in Table 1. The two geometries are sketched in Fig. 1. Sections 3.2–3.10 use this *reference alignment* with $S = 1$, which isolates the effects of geometry, transport and tunnelling at a given barrier. Section 3.5 then uses the measured electron affinity $\chi = 4.6$ eV of bulk InSe [11], which places E_V of the 6-layer film at 5.90 eV, together with the measured and calculated pinning of InSe contacts. Calculated ionisation energies are lower, about 5.2 eV [12], so the absolute alignment is uncertain by several tenths of an eV; results are therefore also given versus the hole barrier ϕ_{Bp}, which is the quantity that matters.

2.2 Carrier transport

Poisson's equation with $\rho = q(p-n+N_D-N_A)$ is solved in InSe and oxide together with Scharfetter–Gummel continuity equations. Fermi–Dirac statistics enter through a degeneracy correction $\Delta(u) =$

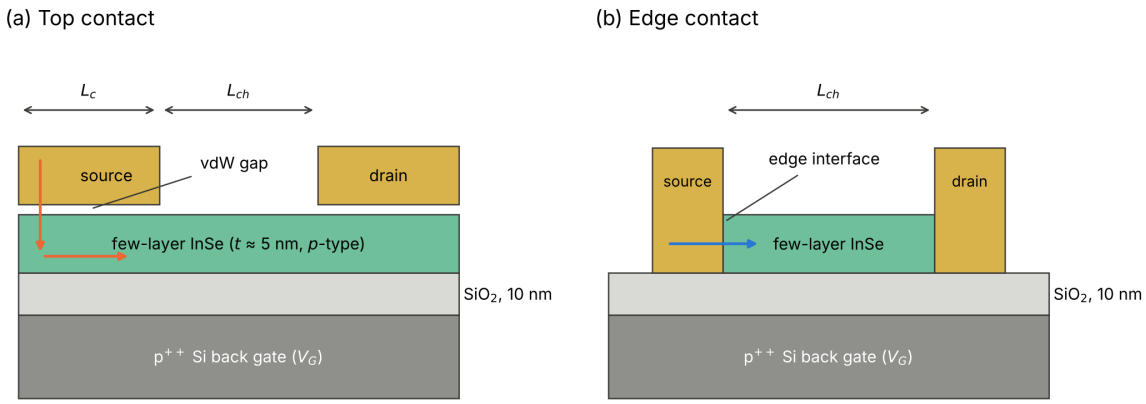


Figure 1: Simulated geometries (not to scale). (a) Top contacts of length L_c separated from the film by a van der Waals gap; holes cross the gap and then flow laterally. (b) Edge contacts abutting the end faces of the film. Both devices sit on 10 nm SiO_2 with a back gate.

Table 1: Parameters (6-layer InSe unless stated). M: measured; L: literature calculation or compilation; R: representative value within a measured range; A: assumed. Sensitivities are given in Sec. 3.10.

Parameter	Value	Source / type
$E_g(N)$, $E_V(N)$	$1.26 + 2.78N^{-1.7}$ eV; 1.39, 5.00 eV at $N = 6$	fit to PL [13, 14] (M); 32% VB share (A)
m_e, m_h (DOS and tunnelling)	0.07, 0.17 m_0	L [15]
ε_r	6.4	L
$\mu_{n,\parallel}$, $\mu_{p,\parallel}$	1000, 50 cm^2/Vs	M [1]; A
$\mu_\perp/\mu_\parallel$	0.01 (range 0.002–1)	M (electrons; 0.002 in vertical FETs [9], ~0.01 bulk)
$\mu(T)$	$\propto T^{-1.5}$	A
$\kappa_\parallel$, $\kappa_\perp$	8.5, 0.76 W/mK	M, time-domain thermoreflectance [10]
c_V	1.43 J/cm^3K	Dulong–Petit (L)
N_A	10^{16} cm^{-3}	A
S_{top}, S_{edge}, Φ_{CNL}	1, 1, 4.35 eV (reference); 0.05 and 0.5 measured, 0.18 calculated (Sec. 3.5)	A (varied); M [11]; L [12]
vdW gap d, Φ_{vdW}	0.30 nm, 2.0 eV ($\mathcal{T}_{vdW} = 0.013$)	A (varied)
oxide; R_{TBR}	10 nm SiO_2 (κ=1.4 W/mK); 10^{-8} m^2K/W at the gate	A

$\eta(u) - \ln u$ ($u = p/N_V$; Nilsson inverse, accuracy better than $0.006kT$) added to the SG drift potential, and through $p_{\mathrm{eq}} = N_V F_{1/2}(-\phi_{Bp}/kT)$ at the contacts [16]. Barriers with $\Phi_{\mathrm{eff}} > E_V$ are therefore allowed (degenerate contacts). Mobility is the tensor $\mu(\hat{\mathbf{e}}) = \mu_{\parallel}(\hat{e}_x^2 + r_\mu \hat{e}_y^2)$ along each mesh edge, which is exact on the axis-aligned mesh used. Thermal velocities, N_C, N_V, n_i and the contact injection follow the local lattice temperature.

2.3 Contacts and tunnelling

The effective work function is $\Phi_{\mathrm{eff}} = \Phi_{\mathrm{CNL}} + S(\Phi_M - \Phi_{\mathrm{CNL}})$, with separate S_{top} and S_{edge}. Carriers at the interface obey the flux condition $J_{p,s} = q v_R \mathcal{T}_{\mathrm{vdW}}(p_s - p_{\mathrm{eq}})$, with $v_R = A^* T^2/qN_V$, balanced against the drift-diffusion current; this is the thermionic-diffusion contact of Crowell and Sze [17]. Tunnelling below the barrier top is treated nonlocally, in the spirit of Ieong *et al.* [18]. For every contact node, the band edge along the inward normal defines the WKB transmission $\mathcal{T}(E)$ to the classical turning point. The Tsu–Esaki current density is

$$J(E)\,dE = \frac{A^* T}{k}\,\mathcal{T}_{\mathrm{vdW}}\,\mathcal{T}(E)\ln\frac{1 + e^{(\epsilon_{F,m} - E)/kT}}{1 + e^{(\epsilon_{F,s}(x_t) - E)/kT}}\,dE, \tag{1}$$

and it is deposited as generation at the turning point. The generation is written semi-implicitly as $G = a - b\,p$ (metal supply minus semiconductor back-flow), so it vanishes identically at equilibrium. It is iterated with under-relaxation until carrier densities change by less than 10^{-3}. Top-contact paths are vertical through the film and also include $\mathcal{T}_{\mathrm{vdW}}$. For comparison, a local field-enhancement model $1 + \Gamma(F)$ of the interface velocity [19] and a no-tunnelling model are also run.

2.4 Heat

The lattice heat equation $c_V \partial_t T = \nabla \cdot (\boldsymbol{\kappa} \nabla T) + H$ has $\boldsymbol{\kappa} = \mathrm{diag}(\kappa_{\parallel}, \kappa_{\perp})$ and

$$H = -\mathbf{J}_p \cdot \nabla \phi_p - \mathbf{J}_n \cdot \nabla \phi_n - T\mathbf{J}_p \cdot \nabla P_p - T\mathbf{J}_n \cdot \nabla P_n, \qquad P_p = \tfrac{k}{q}(2 - \eta_p), \tag{2}$$

which contains Joule heat and Peltier–Thomson heat [20]. Each contact additionally receives the interface Joule heat $J_s \Delta\phi$ and the junction Peltier heat $J_s T P_s$. Tunnelling carriers deposit their energy relative to the local mean carrier energy at the turning point, and the metal side receives the complementary Nottingham-type term. With these terms the total heat equals IV (Sec. 2.6). Heat leaves through the oxide to the gate, which is a heat sink with a thermal boundary resistance R_{TBR}. Metal contacts are adiabatic unless stated otherwise.

2.5 Extraction

$R_{\mathrm{tot}} = V_D/I_D$ is reported per unit width; $1\ \Omega \cdot \mathrm{cm}$ equals $10^4\ \Omega \cdot \mu\mathrm{m}$. It is decomposed exactly into $R_{c,s} + R_{\mathrm{ch}} + R_{c,d}$ using the majority-carrier quasi-Fermi potential on the film centre line at the inner contact edges.

2.6 Verification

Table 2 and Fig. 2 summarise nine tests: (V1) an ohmic resistor; (V2) a finite-velocity contact resistor; (V3) Mott–Gurney space-charge-limited current with Boltzmann and Fermi–Dirac statistics [21]; (V4) thermionic emission at 300 and 350 K; (V5) energy and charge conservation in 2D, gated and ungated (heat out equals IV within 0.3% except 1.2% in the hottest gated case, where Thomson heat from temperature gradients is not separately balanced); (V6) nonlocal tunnelling against

an independent quadrature on the same band profile (0.17%), and on a depletion-approximation profile (13%, the approximation's own error); (V7) a degenerate contact at equilibrium; (V8) transient energy conservation and approach to the DC state; and (V9) Gauss's law with a gate. R_{tot} changes by less than 1.3% between 0.5× and 3× mesh density (Supplementary).

Table 2: Verification summary (`data_v3/verification_summary.csv`).

Test	Quantity	Relative error
V1	R ohmic (Ohm cm^2)	+1.9e-14
V2	R Robin v=1e+09 (Ohm cm^2)	+4.5e-10
V2	R Robin v=1e+05 (Ohm cm^2)	+4.5e-05
V2	R Robin v=1e+04 (Ohm cm^2)	+1.2e-03
V3	boltzmann J/J_MG at 160 V	+7.5e-03
V3	fd J/J_MG at 160 V	-5.5e-03
V4	ideality (300 K)	-6.3e-03
V4	|J_rev(0.3V)| (300 K)	-1.2e-03
V4	ideality (350 K)	-9.4e-03
V4	|J_rev(0.3V)| (350 K)	-1.4e-03
V5	top 4.8eV 0.05V P_out/IV	-2.7e-04
V5	top 4.8eV 0.05V -I_src/I_drn	+1.3e-14
V5	top 5.0eV 0.5V P_out/IV	+2.7e-04
V5	top 5.0eV 0.5V -I_src/I_drn	-2.8e-14
V5	top-gated 4.8eV 0.05V P_out/IV	-4.7e-04
V5	top-gated 4.8eV 0.05V -I_src/I_drn	+0.0e+00
V5	top-gated 5.0eV 0.5V P_out/IV	+2.3e-03
V5	top-gated 5.0eV 0.5V -I_src/I_drn	+5.4e-15
V5	edge 4.8eV 0.05V P_out/IV	+1.4e-03
V5	edge 4.8eV 0.05V -I_src/I_drn	-1.8e-14
V5	edge 5.0eV 0.5V P_out/IV	+2.9e-03
V5	edge 5.0eV 0.5V -I_src/I_drn	+4.4e-16
V5	edge-gated 4.8eV 0.05V P_out/IV	+7.5e-04
V5	edge-gated 4.8eV 0.05V -I_src/I_drn	-1.8e-16
V5	edge-gated 5.0eV 0.5V P_out/IV	+1.2e-02
V5	edge-gated 5.0eV 0.5V -I_src/I_drn	+2.0e-15
V6	J_tun: model vs quad on same profile	-1.7e-03
V6	J_tun: model vs depletion-approximation quadrature (approximation difference)	-1.3e-01
V7	p_s / (N_V F1/2(-phiB/kT)) [exact F]	-4.7e-04
V7	|I(V=0)| (A/cm^2)	4.8e-07 (abs.)
V7	max|phi_p| at equilibrium (V)	1.7e-04 (abs.)
V8	(E_out + E_stored)/E_in (0-1 us)	-4.2e-04
V8	T_max(1 us) - 300 vs DC	+0.0e+00
V9	(sum contact charge + semi charge)/|Q_gate|	-2.5e-06

2.7 Numerical benchmark on Au/MoS_2

No measurements of hole injection into p-InSe exist that could fix the interface parameters. We therefore use a different, well-characterised system only as a numerical benchmark of the solver, contact and tunnelling implementation: the best-characterised metal/2D contact, ultra-high-vacuum Au on 4.5-nm MoS_2 with a 90-nm SiO_2 back gate [22]. The measured quantities at $n = 10^{13}$ cm^{-2} are $R_c = 740\ \Omega\cdot\mu$m, $\rho_c \approx 3\times10^{-7}\ \Omega\,$cm^2 and $L_T \approx 35$–40 nm at 300 K, and $\rho_c \approx 1.5\times10^{-7}\ \Omega\,$cm^2

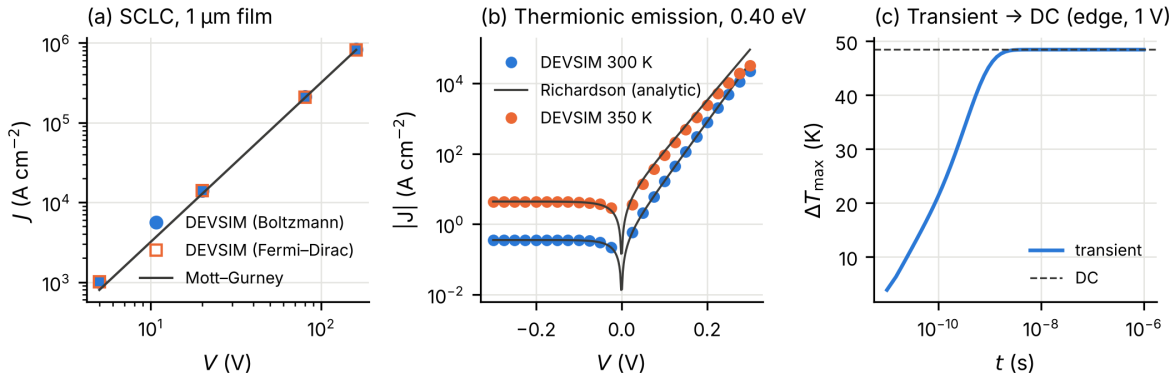


Figure 2: Verification. (a) Mott–Gurney law with Boltzmann and Fermi–Dirac statistics. (b) Thermionic emission at 300 and 350 K. (c) Transient heating of a gated edge device stepped to 1 V (tunnelling off), approaching the DC solution.

and $L_T \approx 20$ nm at 400 K. We keep the 0.15-eV electron barrier assumed in that work and use literature MoS_2 parameters, with an under-contact mobility of 35 cm^2/Vs scaling as $T^{-1.6}$. We fit a single parameter, $\mathcal{T}_{vdW}$, to $R_c(300\,K)$ only. The quantities predicted out of sample are then $L_T(300\,K)$ (from simulated $R_c(L_c)$ with $L_c = 20$–200 nm), and ρ_c and L_T at 400 K.

3 Results

3.1 Benchmark and out-of-sample test

The fit gives $\mathcal{T}_{vdW} = 1.0$, at its physical bound: $R_c = 731\ \Omega \cdot \mu m$ against $740\ \Omega \cdot \mu m$ measured. With the literature barrier, nonlocal tunnelling carries 99.8% of the injection, and no additional vdW tunnel barrier is needed. This is consistent with the clean interfaces of that study. Without further adjustment the model gives $L_T = 34$ nm and $\rho_c = 2.7 \times 10^{-7}\ \Omega\,cm^2$ at 300 K, both within the experimental range. At 400 K it predicts $\rho_c = 2.0 \times 10^{-7}\ \Omega\,cm^2$ (measured 1.5) and $L_T = 29$ nm (measured ≈20, quoted uncertainty up to 50%), with the measured direction of change for both (Fig. 3). The model underestimates how much L_T shrinks with temperature. This is consistent with its temperature-independent barrier and no phonon-assisted injection. We take the agreement as validation of the contact and tunnelling implementation at the 10–40% level. It is a check of the numerics only: MoS_2 has a different band structure, no strong anisotropy and no layer-dependent gap, so it does not validate the InSe parameters or the InSe-specific results.

3.2 Ungated InSe: work function, statistics and tunnelling

Without a gate (Fig. 4), R_{tot} at 0.05 V falls exponentially with Φ_M down to $E_V = 5.0$ eV. For 4.8 eV it is 1430 $\Omega \cdot$ cm (edge), 5080 $\Omega \cdot$ cm (top, 50 nm) and 11 500 $\Omega \cdot$ cm (top, 10 nm). With Fermi–Dirac statistics the resistance keeps decreasing for degenerate contacts: at 5.65 eV it is 5.4, 7.3 and 7.7 $\Omega \cdot$ cm, a factor 2.1–2.6 below the Boltzmann result, which is clipped at E_V. Contacts carry 67% (top, 50 nm) and 87% (top, 10 nm) of R_{tot} at 4.8 eV, against 7.5% for the edge. In the ungated, fully depleted film, nonlocal tunnelling carries 44% of the edge-contact current and lowers its resistance by 13% relative to pure thermionic emission. For top contacts it carries ≤8.5% and lowers resistance by 2–6%. For the edge contact the local $\Gamma(F)$ model recovers only a fifth of this

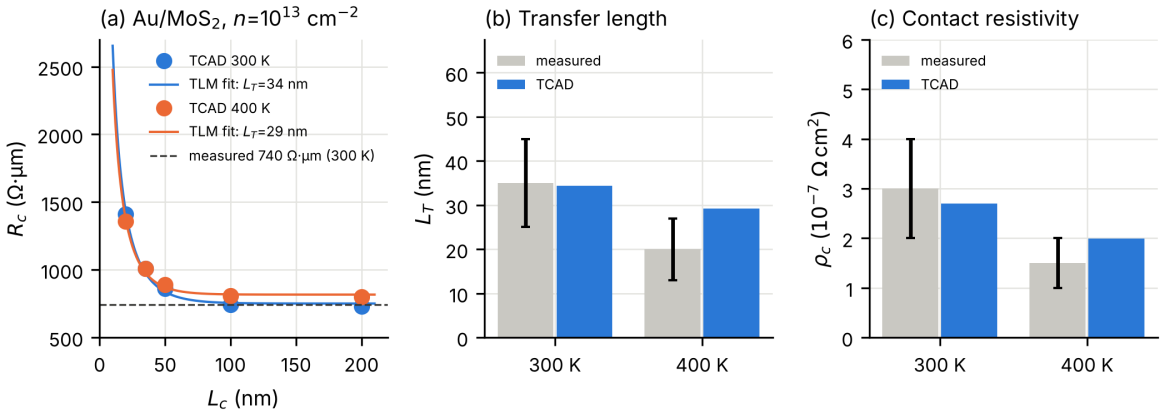


Figure 3: (a) Simulated $R_c(L_c)$ of Au/MoS_2 at 300 and 400 K with transmission-line fits. (b) Transfer length and (c) specific contact resistivity against the measurements of Ref. [22], with their quoted uncertainties. Only R_c at 300 K was fitted; the 400 K values and L_T are predictions.

reduction (2.3% vs 12.6%), because only nonlocal injection bypasses the diffusion-limited part of the barrier; for top contacts, where injection is limited by the interface, the two models differ by about 1%.

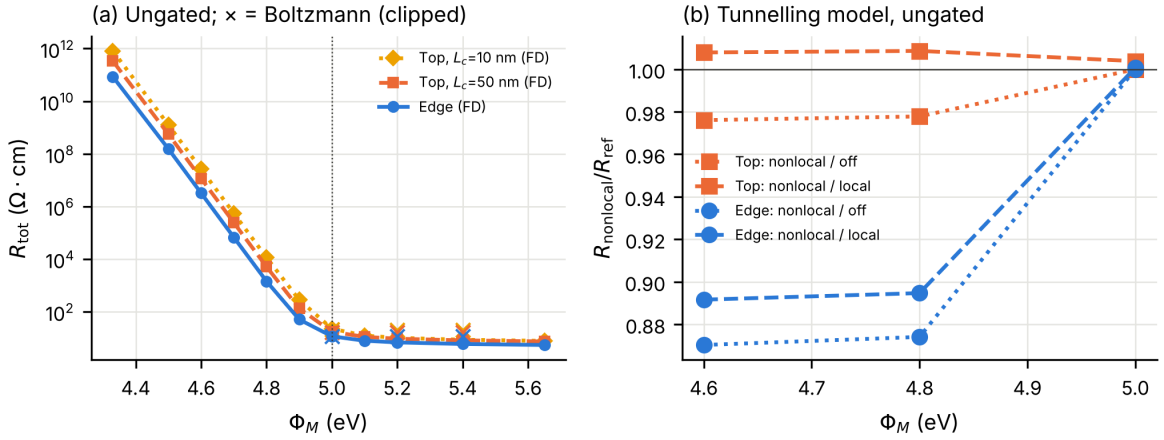


Figure 4: Ungated devices. (a) R_{tot} versus Φ_M with Fermi–Dirac statistics (lines) and Boltzmann statistics clipped at E_V (×). (b) Ratio of nonlocal-tunnelling resistance to the no-tunnelling and local-Γ models.

3.3 Anisotropy and top-contact length

Out-of-plane transport limits top contacts (Fig. 5). Lowering $\mu_\perp/\mu_\parallel$ from 1 to 0.002 doubles the top-contact resistance at 4.8 eV (3920→7990 Ω · cm for 50 nm; 8270→16 700 Ω · cm for 10 nm), whereas edge contacts are unaffected. The edge advantage at 4.8 eV therefore grows from 64% (isotropic) to 72% (baseline) and 82% ($r_\mu = 0.002$). The top-contact resistance saturates for $L_c \gtrsim 50$–100 nm, which indicates transfer lengths of a few tens of nanometres. Thermal anisotropy

matters little at this thickness: with $\kappa_\perp = 0.76$ instead of 8.5 W/mK, the temperature rise of ungated devices at 1 V changes by less than 4%.

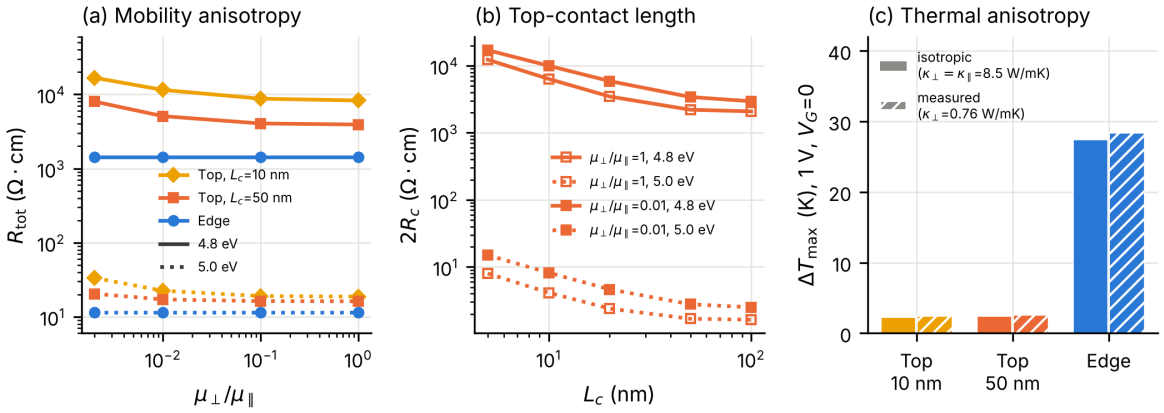


Figure 5: (a) R_{tot} versus out-of-plane/in-plane mobility ratio (colour: geometry; line style: Φ_M). (b) $2R_c$ of top contacts versus contact length for two mobility ratios. (c) Peak temperature rise of ungated devices at 1 V for isotropic and measured $\kappa_\perp$.

3.4 Pinning: can top contacts win?

Edge contacts expose dangling bonds and may be more strongly pinned than vdW top contacts. Figure 6 varies S_{edge} with $S_{\mathrm{top}} = 1$. Already at $S_{\mathrm{edge}} = 0.7$ the edge contact is 180× more resistive than with $S_{\mathrm{edge}} = 1$ at 4.8 eV (2.6×10^5 Ω · cm), and 50× worse than the 50-nm top contact. It overtakes the top contact again only for metals with $\Phi_M \geq 5.4$ eV. The geometric and vdW advantage of edge contacts is therefore smaller than the effect of a modest difference in pinning. Which geometry wins is decided by interface chemistry, and S_{edge} is the quantity experiments should measure first.

3.5 Measured pinning and literature band alignment

Fermi-level pinning has been measured for InSe. Chen *et al.* [11] extracted barriers for In, Ti, Cr and Pd contacts and found $S = 0.05 \pm 0.02$ for direct metal/InSe contacts, with the pinning level about 0.3 eV below E_C; a one-layer oxide interlayer raised S to 0.5. Hybrid-functional calculations for ideal, defect-free interfaces give $S = 0.18$ with metal-induced gap states pinning E_F about 0.58 eV above E_V [12]. We simulated Ti, Cr, Au, Pd and Pt (4.33–5.65 eV) for these three cases and for $S = 1$, with $\chi = 4.6$ eV and $S_{\mathrm{top}} = S_{\mathrm{edge}}$, and swept the hole barrier directly from 0 to 1 eV (Fig. 7). The on state is $V_G = -3$ V at $V_D = 0.05$ V.

With the measured direct-contact pinning every metal gives $\phi_{Bp} = 1.05$–1.12 eV and an electron barrier of only about 0.3 eV. Direct metal contacts to InSe are therefore electron-injecting, which agrees with the n-type behaviour of most InSe transistors, and an ungated device conducts electrons rather than holes. Under a negative gate, holes are injected almost entirely by tunnelling (≥99.9%), and the edge on-state resistance is 5.2×10^3 Ω · cm for Pt and 1.2×10^4 Ω · cm for Ti, only 2.4× apart for a 1.3-eV range of work function. The choice of metal hardly matters.

With the calculated ideal pinning the barriers fall to 0.52–0.76 eV and the edge on-state resistance to 7.1 Ω·cm (Pt) to 123 Ω·cm (Ti). The measured oxide interlayer gives the same 0.52 eV for

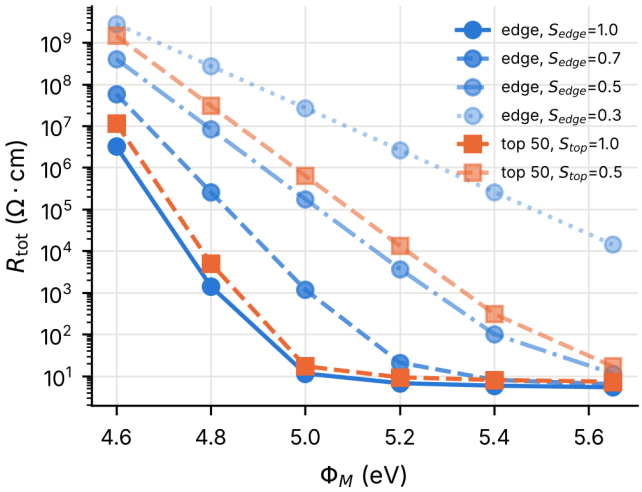


Figure 6: R_{tot} of edge contacts for $S_{\mathrm{edge}} = 1$–0.3 and of 50-nm top contacts for $S_{\mathrm{top}} = 1$ and 0.5 (ungated, 0.05 V).

Pt, a 700× improvement over the direct Pt contact, but 0.79 eV for Pd and Au. Only an unpinned Pt contact reaches 0.25 eV and $R_{\mathrm{tot}} = 0.51\ \Omega\cdot\mathrm{cm}$. Lowering the barrier, by depinning the interface or by choosing a high-work-function metal once it is depinned, is therefore the first lever for p-type InSe; geometry is the second.

The geometry lever becomes stronger, not weaker, as the barrier rises (Fig. 7b). The ratio of top- to edge-contact on-state resistance is 4.1 at $\phi_{Bp} = 0$, 14 at 0.2 eV, 35 at 0.4 eV, 51 at 0.6 eV, 74 at 0.8 eV and 150 at 1.0 eV. At high barriers the current must tunnel, and the edge contact tunnels in-plane with the gate field, while the top contact must tunnel vertically through the vdW gap and the slow out-of-plane path. Without tunnelling the resistance rises by 5×10^2 (edge) and 3×10^2 (top) at 0.2 eV and by 5×10^7 and 8×10^6 at 0.6 eV, so a thermionic contact model would declare any InSe hole contact unusable. Gated runs that did not converge correspond to negligible hole current and are omitted: top contacts with $\phi_{Bp} \geq 1.07$ eV (direct pinning except Pt; Ti and Cr with $S \geq 0.5$) and runs without tunnelling at 0.8–1.0 eV. The ungated resistance is not monotonic above $\phi_{Bp} \approx 0.7$ eV because electron injection takes over.

3.6 Gated Schottky-barrier p-FETs

With the gate (Fig. 8), both geometries switch with on/off ratios above 10^{12} between $V_G = +1$ and -4 V. At $\Phi_M = 4.8$ eV the on-current at $V_D = 0.05$ V and $V_G = -4$ V is 0.21 A/cm (edge) and 0.014 A/cm (top, 50 nm). Practically all of it is injected by tunnelling through the gate-thinned barrier: the no-tunnelling model gives 840× (edge) and 450× (top) less current, and the local $\Gamma(F)$ model 60× and 55× less. At 5.0 eV tunnelling still carries 80–83% of the on-current; at 5.2 eV, where the contacts are degenerate, only 1–16%. At 5.0 eV and $V_G = -4$ V the contact resistance is $R_c = 86\ \Omega\cdot\mu\mathrm{m}$ (edge) against $2080\ \Omega\cdot\mu\mathrm{m}$ (top), a factor of 24, while the channel resistances are nearly equal. The contact therefore limits the top-contacted transistor, and the channel limits the edge-contacted one.

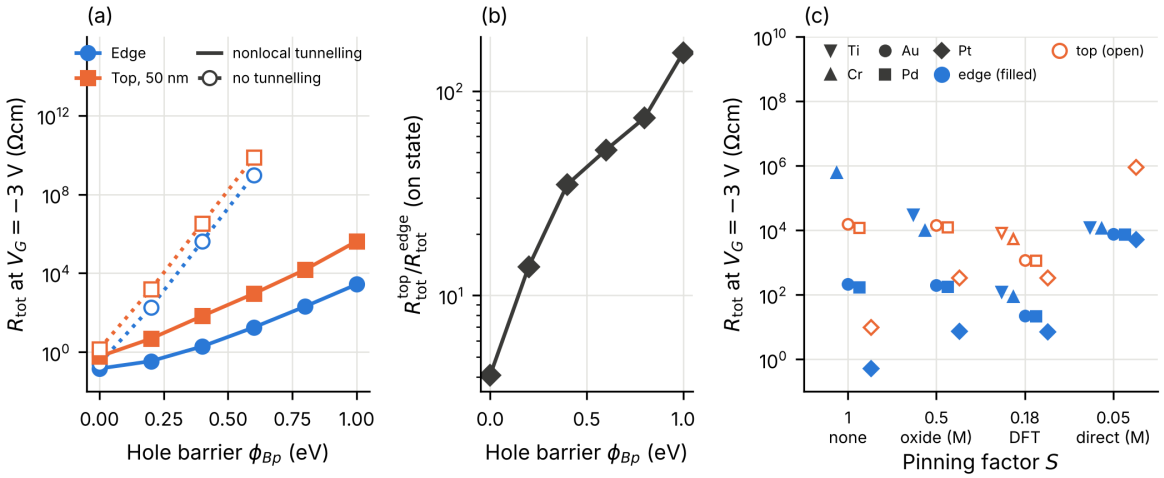


Figure 7: Literature band alignment ($\chi = 4.6$ eV, $E_V = 5.90$ eV for 6 layers). (a) On-state R_{tot} ($V_G = -3$ V, $V_D = 0.05$ V) versus hole barrier, with nonlocal tunnelling (solid) and without tunnelling (dotted). (b) Ratio of top- to edge-contact on-state resistance. (c) On-state R_{tot} for five metals and four pinning cases: no pinning; measured oxide interlayer and direct contact (M) [11]; calculated ideal interface [12]. Filled: edge; open: top, 50 nm. Missing points did not converge and carry negligible hole current.

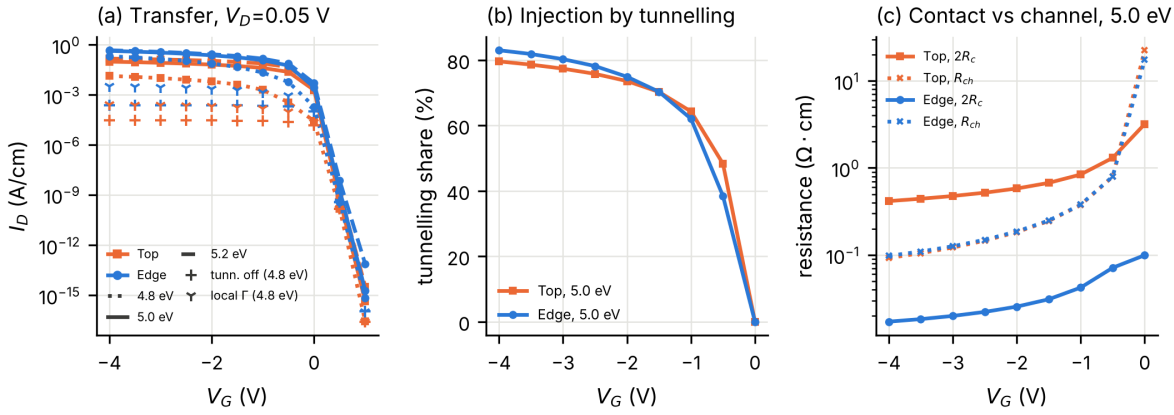


Figure 8: Gated devices (10 nm SiO_2, $V_D = 0.05$ V). (a) Transfer curves for the full nonlocal model (lines), with the no-tunnelling (+) and local-Γ (Y-shaped symbols) models at 4.8 eV. (b) Share of the current injected by tunnelling. (c) Contact and channel resistance at 5.0 eV.

3.7 Thickness

Thinner films have larger band gaps and deeper valence bands (Fig. 9a), so ungated resistances rise steeply below six layers. At 4.8 eV the edge-contact resistance rises from 242 $\Omega \cdot$cm (12 layers) to 1.1×10^5 $\Omega \cdot$cm (3 layers). In the gated on state ($\Phi_M = 5.0$ eV, $V_G = -3$ V) the edge contact improves monotonically with thickness (0.197→0.135 $\Omega \cdot$cm from 3 to 12 layers). The top contact instead has an optimum near four layers: 0.53 $\Omega \cdot$cm, rising to 1.55 $\Omega \cdot$cm at 12 layers. Beyond that thickness the vertical access through layers with low $\mu_\perp$ outweighs the shallower barrier (Fig. 9b).

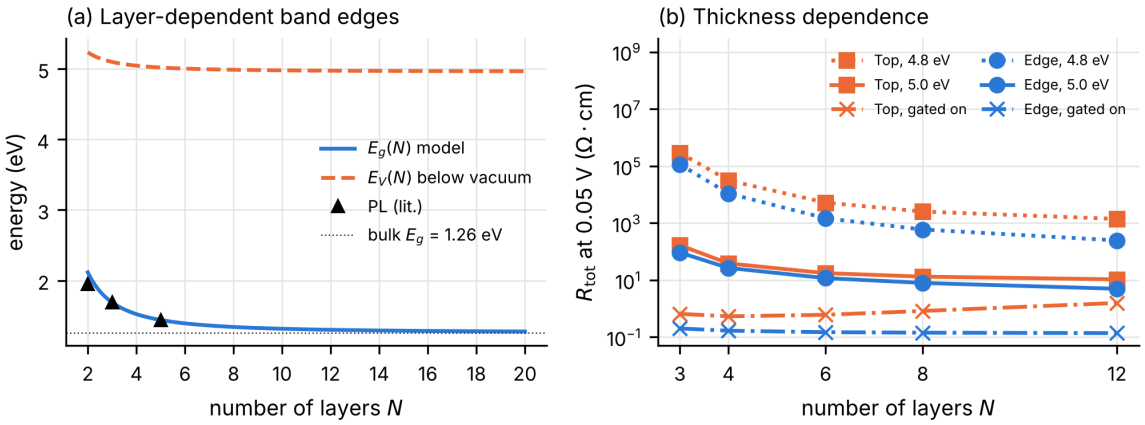


Figure 9: (a) Layer-dependent band gap and valence-band edge with photoluminescence data. (b) R_{tot} versus layer number, ungated and in the gated on state ($V_G = -3$ V, $\Phi_M = 5.0$ eV).

3.8 Self-heating

Table 3 compares the geometries at the same gated operating point ($\Phi_M = 5.0$ eV, $V_G = -2$ V). At $V_D = 1$ V the edge-contacted device carries 3.7 A/cm and heats by 150 K; the 50-nm top contact carries 1.16 A/cm and heats by 24 K. Per unit power the edge device is also about twice as hot (41 vs 21 K cm/W), because its footprint on the oxide is a third of that of the top-contacted film. In the on state at $V_G = -3$ V (Fig. 10a) the edge device reaches about 490 K at 1 V, so its usable bias is thermally limited, not contact-limited. The two-dimensional maps in Fig. 11 show where the heat sits: in the top-contacted device the hottest point lies at the drain-side channel end on the contact side of the film, and the whole film stays within 24 K of ambient; in the edge-contacted device the film is 126–150 K hot everywhere and is coolest at the oxide interface, through which heat leaves to the substrate.

The individual heat terms matter at the 5–40% level (Fig. 10b):

- Using bulk Joule heat only underestimates ΔT_{max} by 7–8% and misses 8–15% of the dissipated power.
- Measured out-of-plane conductivity, compared with isotropic κ, raises ΔT_{max} by 8% (edge) and 17% (top).
- Heat sinking through the metal contacts lowers ΔT_{max} by 23% (edge) and 42% (top).
- A ten times smaller thermal boundary resistance lowers it by 35–37%.

Peltier heat changes $\Delta T_{\max}$ by only about 1 K but redistributes heat: it cools the hole-injecting contact and heats the collecting one (top, 1 V: source +11.9 K, drain +21.6 K with Peltier, against +13.5 and +20.9 K without). Temperature-dependent statistics change temperatures by less than 1%. The lattice responds to a drain step within a few hundred picoseconds, with 63% of the final rise at 0.20 ns (top) and 0.25 ns (edge) (Fig. 10c; this transient was simulated without tunnelling, so its amplitudes are smaller than in Table 3). For pulses longer than a nanosecond the DC temperatures therefore apply.

Table 3: Gated operating points ($\Phi_M = 5.0$ eV, $V_G = -2$ V, baseline thermal model). All quantities in a row refer to the same bias.

Geometry	V_D (V)	I_D (A/cm)	IV (W/cm)	$\Delta T_{\max}$ (K)	ΔT_{source} (K)	ΔT_{drain} (K)
Top, 50 nm	0.5	0.618	0.309	5.9	3.5	5.6
Top, 50 nm	1.0	1.164	1.164	24.0	11.9	21.6
Edge	0.5	2.228	1.114	47.0	40.8	46.2
Edge	1.0	3.693	3.693	149.9	144.4	144.6

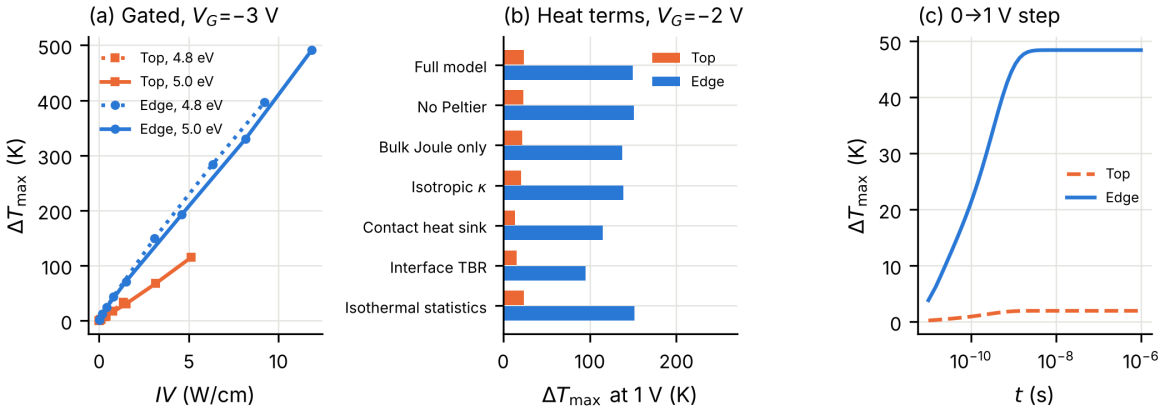


Figure 10: (a) Peak temperature rise versus dissipated power, gated on state. (b) Effect of each heat-model term at $V_D = 1$ V, $V_G = -2$ V, $\Phi_M = 5.0$ eV. (c) Response to a 0→1 V drain step at $V_G = -2$ V, simulated without tunnelling (steady-state rises of 48 K and 2 K); only the time constant is used in the text.

3.9 Compact model

For top contacts in the barrier-limited regime we compare a transmission-line model with the simulation. The model uses

- an interface resistivity $\rho_{\text{if}} = V_t/(q v_R \mathcal{T}_{\text{vdW}} p_{\text{eq}})$;
- a vertical access term $t/(2q\mu_\perp \bar{p})$;
- an under-contact sheet resistance $1/(q\mu_\parallel \bar{p} t)$;
- $R_c = \sqrt{\rho R_{\text{sh}}}\coth(L_c/L_T)$.

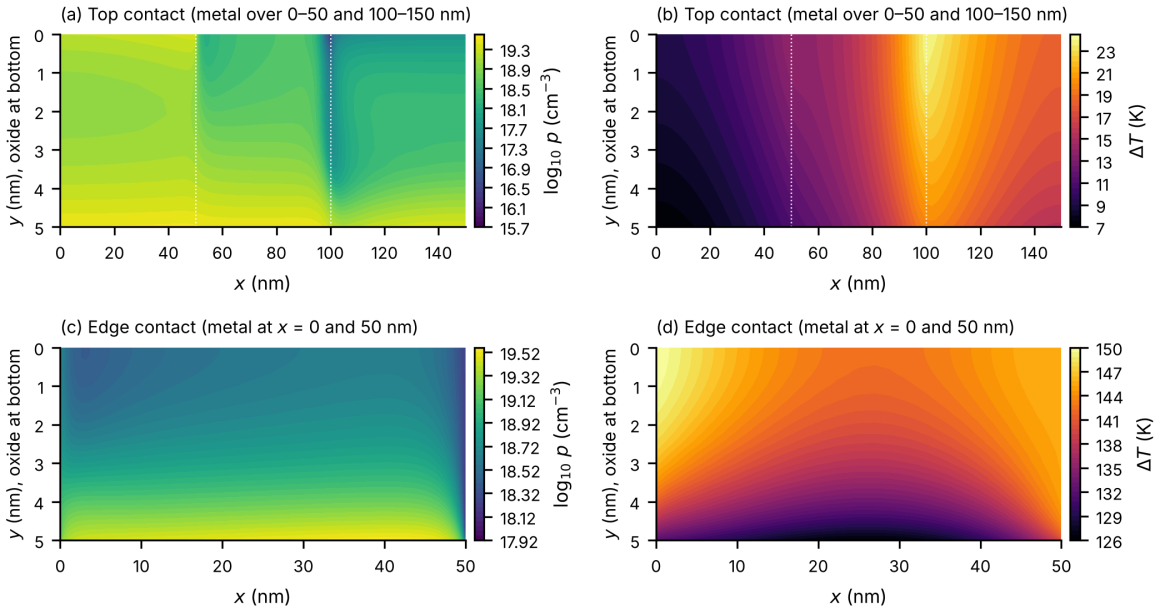


Figure 11: Hole density and temperature rise of gated devices at $V_D = 1$ V, $V_G = -2$ V, $\Phi_M = 5.0$ eV (full model). The vertical axis is stretched: the film is 5 nm thick. The oxide interface is at the bottom ($y = 5$ nm). Dotted lines in (a, b) mark the channel ends.

Over 120 cases ($\phi_{Bp} = 0.05$–0.25 eV, $d_{\mathrm{vdW}} = 0.2$–0.4 nm, $L_c = 10$–100 nm, $r_\mu = 1$ and 0.01) the fully predictive version, with $\bar{p} = p_{\mathrm{eq}}$, has a median error of 11% and a 90th-percentile error of 33% (Fig. 12). Taking $\bar{p}$ from the simulation reduces the 90th percentile to 20%. The compact model is therefore adequate for circuit-level estimates when tunnelling is weak, that is, without strong gating.

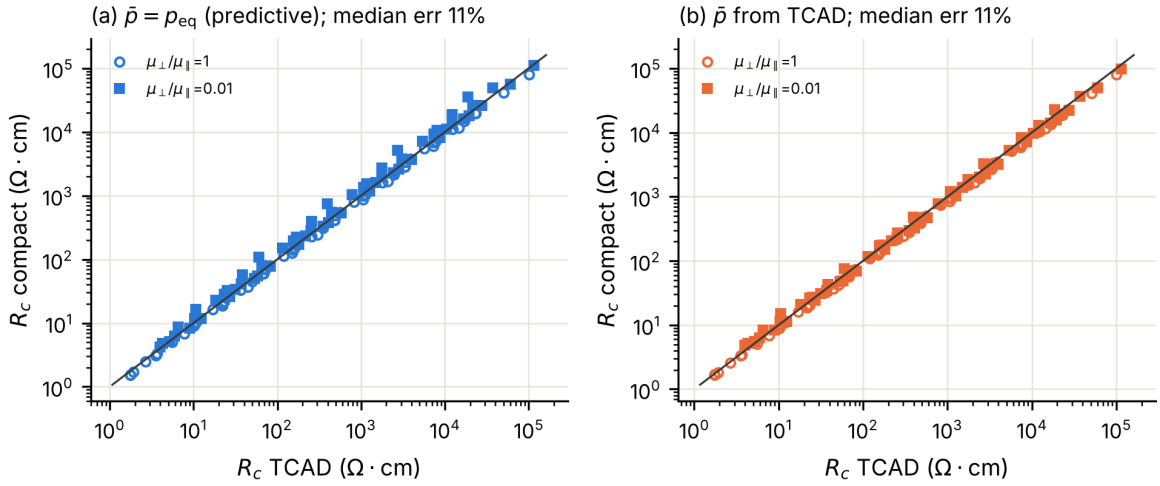


Figure 12: Compact transmission-line model against simulated top-contact R_c (no tunnelling), with (a) $\bar{p} = p_{\mathrm{eq}}$ and (b) $\bar{p}$ from the simulation.

3.10 Uncertainty quantification

We propagated the uncertainty of the least-known inputs by Monte Carlo sampling: 120 samples, each simulated for both geometries at 4.8 and 5.0 eV, ungated, at 0.05 V. The sampling ranges were:

- $\Delta E_V \sim \mathcal{N}(0, 0.1\ \text{eV})$ and $S \sim U(0.3, 1)$, the same for both contacts;
- $d_{\text{vdW}} \sim U(0.2, 0.4)$ nm and $\Phi_{\text{vdW}} \sim U(1, 3)$ eV;
- log-uniform μ_p (20–100 cm^2/Vs), $\mu_\perp/\mu_\parallel$ (0.002–0.1) and N_A (10^{15}–10^{17} cm^{-3});
- $\varepsilon_r \sim U(5, 8)$ and $m_h \sim U(0.1, 0.3)$.

The resulting 5–95% range of R_{tot} spans seven to eight orders of magnitude: $61\text{–}1.2 \times 10^9\ \Omega\cdot\text{cm}$ for the edge contact at 4.8 eV (Fig. 13a). The band alignment (ΔE_V, Spearman $\rho = -0.63$ to -0.77) and the pinning factor ($\rho = -0.69$ to -0.81) account for nearly all of the variance. Their first-order indices are 0.47–0.77, while all other parameters stay at or near the estimator's noise floor of about 0.06–0.17 for $n = 120$ (Fig. 13b). The *ranking* of the geometries is robust: when both contacts share the same pinning, the edge contact has lower resistance in every sample. The edge/top ratio has a median of 0.24, with a 5–95% range of 0.07–0.51 at 4.8 eV and 0.08–0.72 at 5.0 eV. Absolute resistances, by contrast, cannot be predicted to better than several orders of magnitude from these ranges alone. Section 3.5 narrows them using the measured pinning of InSe contacts; the remaining uncertainty is the absolute band alignment and the pinning of edge contacts.

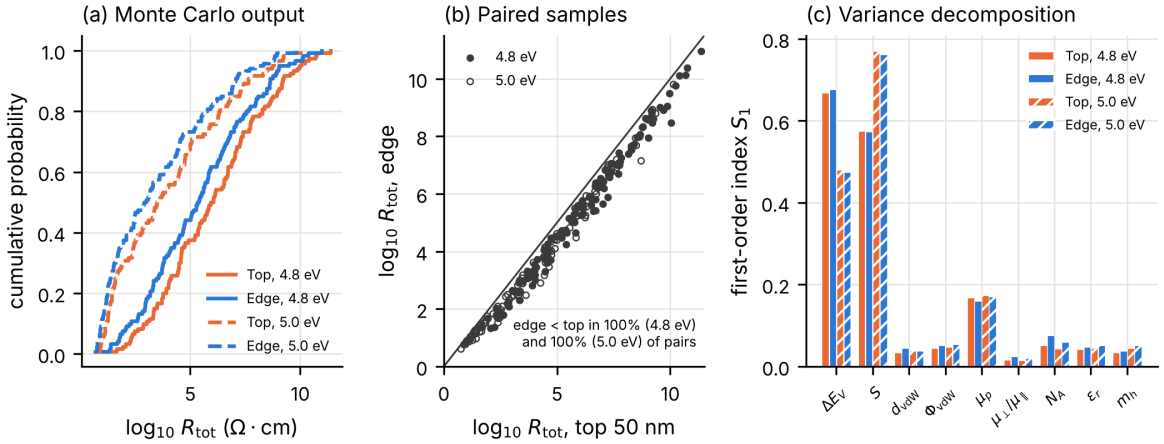


Figure 13: Monte Carlo propagation of parameter uncertainty (ungated, 0.05 V). (a) Cumulative distributions of $\log_{10} R_{\text{tot}}$. (b) Paired samples: the same parameter draw simulated for the top (50 nm) and edge contact; points below the diagonal have lower edge resistance. (c) First-order variance indices (colour: geometry; hatching: 5.0 eV).

4 Conclusions

For hole injection into few-layer InSe p-FETs the model gives five design rules:

1. With the measured pinning of direct metal/InSe contacts ($S = 0.05$), every common metal gives a hole barrier of about 1.1 eV and an electron-injecting contact. Hole operation then relies entirely on gate-induced tunnelling, and the metal work function barely matters (2.4× from Ti to Pt).

2. The barrier must first be lowered by depinning: the measured oxide interlayer brings Pt to 0.52 eV and lowers the on-state resistance 700×.

3. At any barrier, edge contacts beat 50-nm top contacts in the on state, by 14× at 0.2 eV and 150× at 1 eV, because they avoid the vdW gap and the slow out-of-plane path. This holds only if edge contacts are not more strongly pinned than top contacts, which is the one interface property still to be measured.

4. Nonlocal tunnelling is essential: thermionic and local enhancement models underestimate the on-current by orders of magnitude.

5. Top contacts have an optimum film thickness of about four layers; edge-contacted devices carry more current and need heat removal.

The model is verified, benchmarked on Au/MoS_2 with one fitted parameter, anchored to the measured pinning of InSe contacts, and accompanied by a compact model and uncertainty bounds. Its main remaining limitations are the uncertain absolute band alignment of few-layer InSe and the unmeasured pinning of edge contacts.

Statements and Declarations

Author contributions H.A. designed the study, developed the code, performed the simulations and analysis, and wrote the manuscript.

Funding None.

Competing interests None.

Code and data availability https://github.com/Brukrish2006/InSe-Top-Edge-Contact-Resistace (`src_v3`, `data_v3`, `figures_v3`).

References

[1] D.A. Bandurin *et al.*, Nat. Nanotechnol. **12**, 223 (2017), doi: 10.1038/nnano.2016.242.

[2] J. Jiang, L. Xu, C. Qiu and L.-M. Peng, Nature **616**, 470 (2023), doi: 10.1038/s41586-023-05819-w.

[3] B. Qin *et al.*, Science **389**, 299 (2025), doi: 10.1126/science.adu3803.

[4] S. Song *et al.*, Nat. Rev. Electr. Eng. (2025), doi: 10.1038/s44287-025-00251-w.

[5] A. Allain, J. Kang, K. Banerjee and A. Kis, Nat. Mater. **14**, 1195 (2015), doi: 10.1038/nmat4452.

[6] J. Kang, W. Liu, D. Sarkar, D. Jena and K. Banerjee, Phys. Rev. X **4**, 031005 (2014).

[7] L. Wang *et al.*, Science **342**, 614 (2013), doi: 10.1126/science.1244358.

[8] M.H.D. Guimarães *et al.*, ACS Nano **10**, 6392 (2016), doi: 10.1021/acsnano.6b02879.

[9] V.K. Sangwan *et al.*, *Probing vertical conduction in InSe vertical field-effect transistors*, Appl. Phys. Lett. **115**, 243104 (2019), doi: 10.1063/1.5128808.

[10] A. Rai, V.K. Sangwan, J.T. Gish, M.C. Hersam and D.G. Cahill, *Anisotropic thermal conductivity of layered indium selenide*, Appl. Phys. Lett. **118**, 073101 (2021).

[11] Y.-H. Chen *et al.*, *Oxidized-monolayer tunneling barrier for strong Fermi-level depinning in layered InSe transistors*, npj 2D Mater. Appl. **3**, 49 (2019), doi: 10.1038/s41699-019-0133-3.

[12] Y. Guo and J. Robertson, *Band structure, band offsets, substitutional doping, and Schottky barriers of bulk and monolayer InSe*, Phys. Rev. Mater. **1**, 044004 (2017), doi: 10.1103/PhysRevMaterials.1.044004.

[13] G.W. Mudd *et al.*, Sci. Rep. **6**, 39619 (2016), doi: 10.1038/srep39619.

[14] G.W. Mudd *et al.*, *Tuning the bandgap of exfoliated InSe nanosheets by quantum confinement*, Adv. Mater. **25**, 5714 (2013).

[15] S. Magorrian, V. Zólyomi and V. Fal'ko, Phys. Rev. B **94**, 245431 (2016), doi: 10.1103/PhysRevB.94.245431.

[16] D. Bednarczyk and J. Bednarczyk, *The approximation of the Fermi–Dirac integral* $F_{1/2}(\eta)$, Phys. Lett. A **64**, 409 (1978).

[17] C.R. Crowell and S.M. Sze, Solid-State Electron. **9**, 1035 (1966).

[18] M. Ieong, P.M. Solomon, S.E. Laux, H.-S.P. Wong and D. Chidambarrao, IEDM Tech. Dig., 733 (1998).

[19] F.A. Padovani and R. Stratton, Solid-State Electron. **9**, 695 (1966), doi: 10.1016/0038-1101(66)90097-9.

[20] G.K. Wachutka, IEEE Trans. Comput.-Aided Des. **9**, 1141 (1990), doi: 10.1109/43.62751.

[21] N.F. Mott and R.W. Gurney, *Electronic Processes in Ionic Crystals*, 2nd ed., Oxford (1940).

[22] C.D. English, G. Shine, V.E. Dorgan, K.C. Saraswat and E. Pop, *Improved contacts to* MoS_2 *transistors by ultra-high vacuum metal deposition*, Nano Lett. **16**, 3824 (2016), doi: 10.1021/acs.nanolett.6b01309.